

Introduction to Glomerular Disease: Histologic Classification and Pathogenesis

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HISTOLOGIC CLASSIFICATION

Glomerular disease has a wide variety of etiologies and clinical presentations (see [Chapter 16](#)). Some glomerular diseases are given the generic title of glomerulonephritis (GN), which implies an immune or inflammatory pathogenesis. Although a specific diagnosis is sometimes possible based on clinical presentation and laboratory tests, in most patients a kidney biopsy is useful for both classification and prognosis. Ideally, the kidney biopsy should be examined by light microscopy, immunohistology or immunofluorescence, and electron microscopy (EM), which allows description of the underlying histopathological pattern. Using this approach, an etiologic classification or specific diagnosis can often be made, particularly when integrating the biopsy results with clinical features and other laboratory tests. One useful example of an etiologic classification is the Mayo/Renal Pathology Society consensus report on the classification of GN,¹ distinguishing five major disease etiologies: immune-complex GNs, pauci-immune GNs, anti-glomerular basement membrane (GBM) antibody GNs, monoclonal gammopathy-associated GNs, and complement-related GNs. In some cases the exact condition might remain idiopathic, but in the vast majority of cases one of these GN classes can be assigned. However, because treatments are often developed for specific histologic patterns, a histology-based approach is currently favored in the management of most patients with glomerular disorders.

HISTOPATHOLOGY

The full assessment of a kidney biopsy requires light microscopy, EM, and examination for deposits of complement and immunoglobulin by immunofluorescence (IF) or immunoperoxidase (IP) techniques.

Light Microscopy

In GN the dominant, but not the only, histologic lesions are in glomeruli ([Fig. 17.1](#)). GN is described as *focal* (<50% of glomeruli are involved) or *diffuse* (>50% of glomeruli are involved). In any individual glomerulus, injury may be *segmental* (affecting <50% of any glomerulus) or *global* (>50%). Sampling error is possible in a kidney biopsy; the extent of a focal lesion may be misjudged in a small biopsy specimen, and sections through glomeruli may miss segmental lesions. Therefore, the number of glomeruli in a biopsy is an essential quality parameter, and serial sections may be necessary for the biopsy workup. In general and for most lesions and pathologies, 10 to 15 glomeruli are considered optimal and 8 to 10 necessary, albeit in a few cases a diagnosis can be made even on a single glomerulus. Lesions may be hypercellular (termed *proliferative*) because of an increase in endogenous cells or infiltration of inflammatory leukocytes, and based on localization (extracapillary, endocapillary, mesangial or combinations thereof). Severe acute inflammation may produce glomerular necrosis, which is

often segmental. Multiple processes can cause thickening of the walls of the glomerular capillaries, including an increase in GBM material and immune deposits. Segmental sclerosis (scarring) may occur and is characterized by segmental capillary collapse with the accumulation of hyaline material and extracellular matrix, often with attachment of the capillary wall to Bowman's capsule (synechiae or adhesion formation).

The classic stains used in light microscopy include hematoxylin-eosin (HE) and the periodic acid-Schiff (PAS) reaction, the latter being particularly effective for evaluating the overall histological pattern. More specific stains include methenamine silver, which stains GBM and other matrix black and which is particularly helpful in revealing GBM abnormalities (such as double contours or spikes of the GBM), which are not easily detected with other techniques. Trichrome staining is also useful to show areas of scarring (blue) and immune deposits or thrombi (red).

Extracapillary proliferations, or *crescents*, are defined as increased number of cells in Bowman's space. Crescents develop when severe glomerular injury results in local rupture of the capillary wall or Bowman's capsule, allowing plasma proteins and inflammatory material to enter into Bowman's space. Crescents mainly consist of proliferating parietal cells, probably some visceral epithelial cells, and infiltrating fibroblasts; lymphocytes and monocytes/macrophages might be also involved, often with local fibrin deposition. They are called crescents because of their appearance when the glomerulus is cut in one plane for histology. Crescents are destructive, may rapidly increase in size, and may lead to glomerular tuft occlusion (see [Fig. 17.1](#)). If the acute injury is stopped, the crescents may probably either resolve with restitution of normal morphology or, more likely, heal via fibrosis (i.e., via transformation to fibrocellular and later fibrous crescents, which in turn may contribute to irreversible loss of renal function). Crescents are most frequently observed in vasculitis and in Goodpasture disease but also in severe acute GN of any cause.

Tubulointerstitial injury, inflammation, and tubular atrophy and fibrosis can accompany GN, particularly the latter, which is usually strongly correlated with prognosis (see [Chapter 81](#)).

Immunofluorescence and Immunoperoxidase Microscopy

Indirect IF and IP staining are both used to identify immune reactants ([Fig. 17.2](#) and [Table 17.1](#)), particularly for immunoglobulins (IgG, IgA, and IgM), for components of the complement system (usually C3 and C1q in native and C4d in transplant biopsies), and for the presence of fibrin, which is typically observed in crescents and in capillaries in thrombotic disorders such as hemolytic uremic syndrome (HUS) and antiphospholipid syndrome. For diagnostic purposes, not only the composition of the deposits is important but also their localization; that is, in the mesangium or along capillary loops (distinguishing sub-epithelial, subendothelial, or combined localization) and in the staining pattern (i.e., continuous [linear] or discontinuous [granular]).

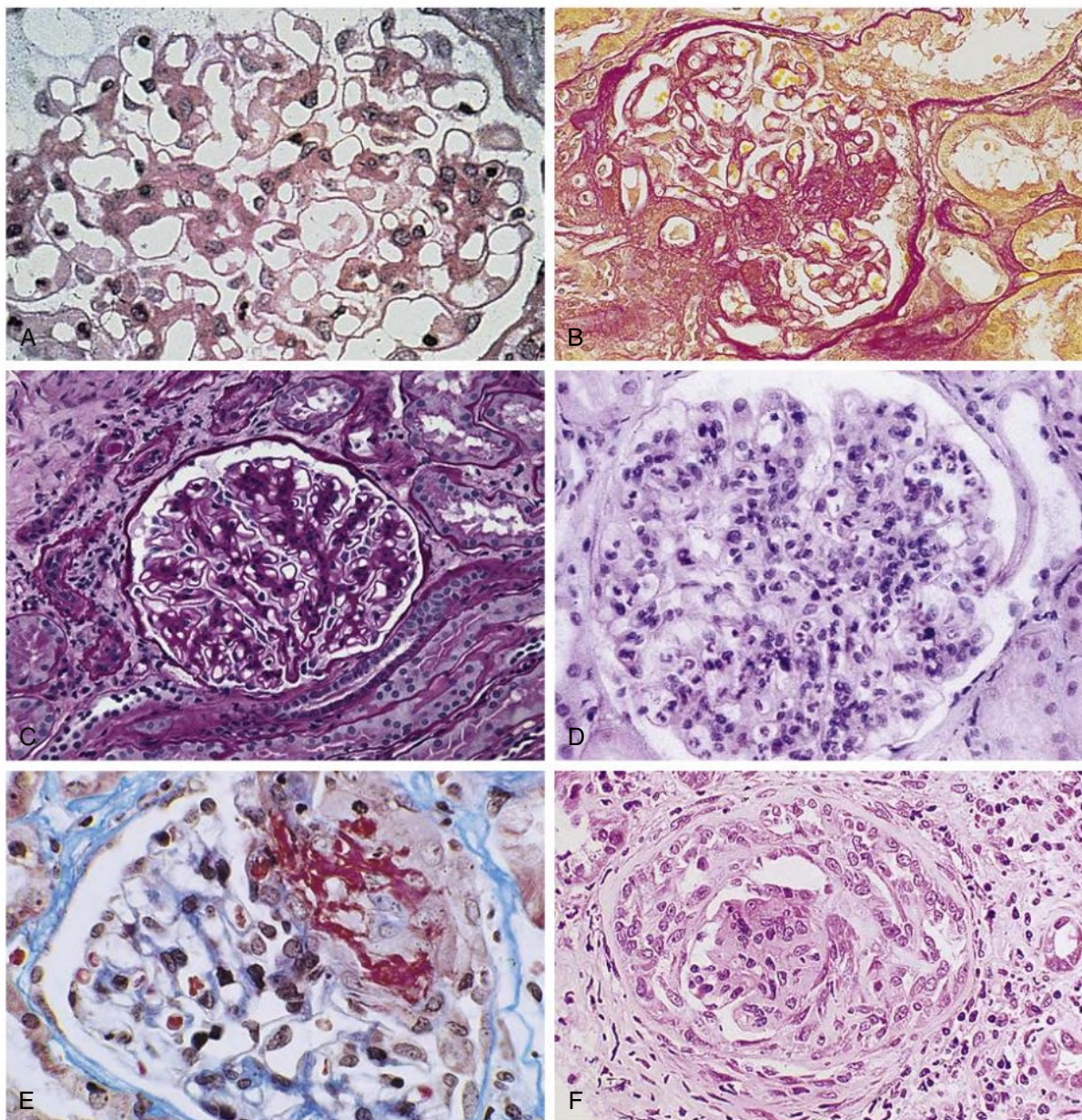


Fig. 17.1 Pathology of Glomerular Disease: Light Microscopy. Characteristic patterns of glomerular disease illustrating the range of histologic appearances and the descriptive terms used. (A) Normal glomerulus: minimal change disease. (B) Segmental sclerosis: focal segmental glomerulosclerosis. (C) Diffuse mesangial hypercellularity: immunoglobulin A nephropathy. (D) Diffuse endocapillary hypercellularity: poststreptococcal glomerulonephritis. (E) Segmental necrosis: renal vasculitis. (F) Crescent formation: anti-glomerular basement membrane disease. (A–B, hematoxylin-eosin; C–D and F, periodic acid–Schiff; E, trichrome.)

Electron Microscopy

Electron microscopy (EM) is valuable for defining the morphology of the GBM, which is abnormal in some forms of hereditary nephropathy (e.g., Alport syndrome and thin basement membrane nephropathy; see [Chapter 48](#)) and for identifying fibrils (e.g., in amyloidosis) or tubuloreticular intracellular structures (e.g., in lupus nephritis). EM is also useful for localizing the site of immune deposits, which are usually homogeneous and electron dense ([Fig. 17.3](#)). Electron-dense deposits are seen in the mesangium or along the capillary wall on the subepithelial or subendothelial side of the GBM. Infrequently, the electron-dense material follows a linear pattern within the GBM. The sites of immune deposits are helpful in the classification of the types of GN.

GENERAL MECHANISMS OF GLOMERULAR INJURY

Proteinuria

Proteinuria, accompanied by variable degrees of hematuria, is the hallmark of glomerular disease. The endothelial glycocalyx and the GBM may repel proteins in part through their highly negative charge (proteins are mostly negatively charged as well) and prevent them from entering Bowman's space. The key barrier for protein is the slit diaphragm between the podocyte foot processes^{2,3} ([Fig. 17.4](#)). The slit diaphragm consists of several transmembrane proteins that extend from adjacent interdigitating foot processes to form a zipper-like scaffold on the outer side of the GBM ([Fig. 20.1](#) and [Chapter 1](#)).

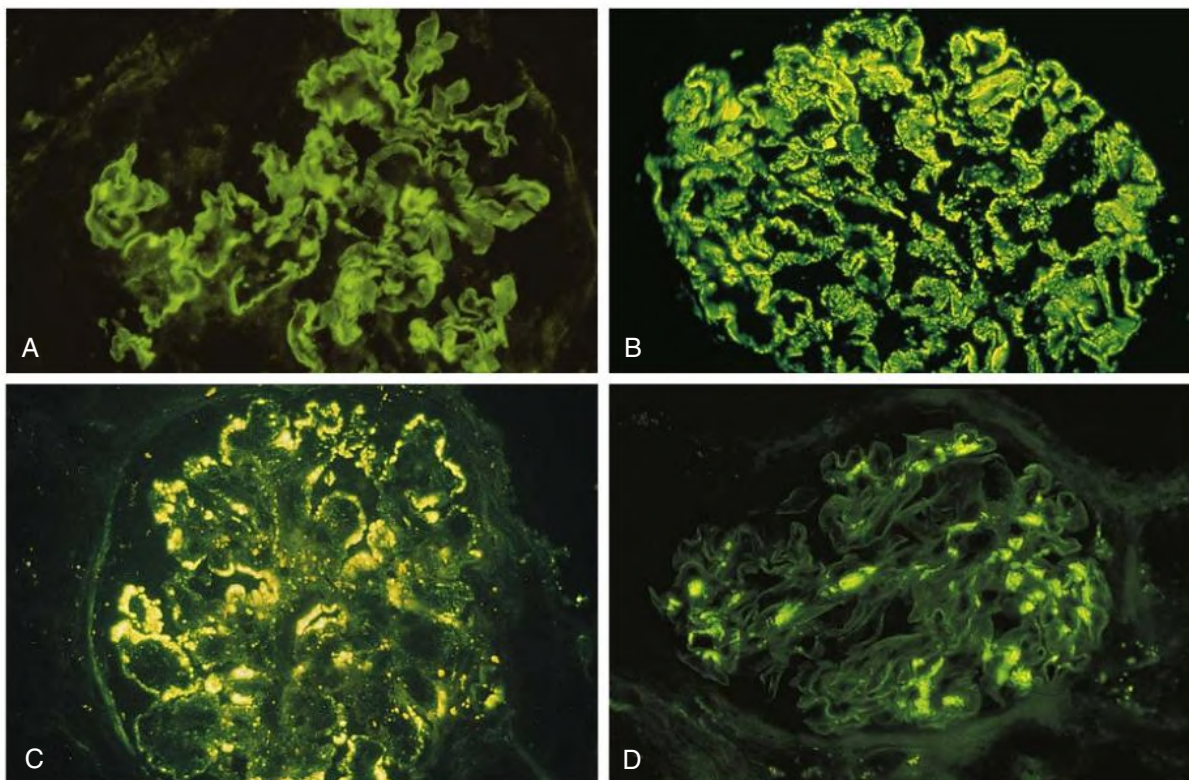


Fig. 17.2 Pathology of Glomerular Disease: Immunofluorescence Microscopy. Common patterns of glomerular staining found by immunofluorescence. (A) Linear capillary wall immunoglobulin (Ig) G: anti-glomerular basement membrane disease. (B) Fine granular capillary wall IgG: membranous nephropathy. (C) Coarse granular capillary wall IgG: membranoproliferative glomerulonephritis type I. (D) Granular mesangial IgA: IgA nephropathy.

The importance of the slit diaphragm in proteinuric states has been documented in numerous hereditary types of nephrotic syndrome in which the mutations involve various slit diaphragm proteins (see [Chapter 20](#)). These diseases usually manifest as a type of steroid-resistant minimal change disease (MCD) or focal segmental glomerulosclerosis (FSGS). Whereas most recessive mutations of slit diaphragm or podocyte proteins manifest in childhood or even prenatally, dominant mutations tend to manifest in early adult life.³ An exception is steroid-resistant autosomal recessive nephrotic syndrome, in which the homozygous mutation in podocin (*NPHS2*) presents in childhood, but the heterozygous mutation, when it coexists with the p.R229Q variant polymorphism, may manifest clinically in young adulthood (age 20–35 years).⁴

Although it may result from injury or mutation of slit diaphragm proteins, proteinuria may be caused by nonspecific injury to the podocyte in many cases. When the podocyte is injured, it may undergo shape changes with swelling and loss or fusion of the foot processes. Filtration is reduced at sites where the foot processes fuse (possibly accounting for the reduction of the filtration coefficient K_f seen in nephrotic syndrome), but there are gaps where the podocytes are detached from the GBM. Massive protein filtration may occur at these sites; structurally, the capillary wall defects are likely to correspond to the large pores noted in functional studies⁵ ([Fig. 17.5](#)). Podocyte immaturity also can result in nephrotic syndrome, perhaps from incomplete differentiation and slit diaphragm development. Congenital nephrotic syndrome with mesangial sclerosis has been linked with mutations in the phospholipase C epsilon gene (*PLCE1*), which is important in podocyte development.⁶

In addition to podocyte damage, and in particular slit diaphragm defects, proteinuria also can result from changes in the glomerular

endothelium, especially its glycocalyx, as well as from changes in the GBM and altered electrical forces across the GBM.

Severe albuminuria reflects a glomerular defect, but some albumin is normally filtered but then endocytosed and metabolized in the proximal tubule or is transcytosed intact through the tubular cell. Proximal tubular dysfunction can therefore result in albuminuria if endocytosis is impaired, although this is generally in the nonnephrotic range.

Antibody and Antigen

Many glomerular diseases are associated with deposition or glomerular trapping of immunoglobulins, often with components of the complement system, and with the presence of electron-dense deposits by EM. These findings likely represent immune complexes. Experimentally, immune complexes can localize in glomeruli by two major mechanisms. In some conditions, such as mesangial proliferative GN, diseases associated with a membranoproliferative GN (MPGN) pattern, or lupus nephritis, the immune complexes are thought to originate in the circulation and to be passively trapped in the mesangium or subendothelial areas. However, circulating immune complexes cannot readily pass across the GBM. Therefore, the presence of immunoglobulin G (IgG) on the subepithelial aspect of the basement membrane, such as occurs in membranous nephropathy (MN), either results from the direct binding of podocyte antigens by an antibody (e.g., phospholipase A2 receptor antibody) or represents binding of an antibody to an antigen that was temporarily “trapped” or bound at this site (in situ complex formation).⁷ GN also may occur only with complement activation in the glomeruli in the absence of IgG, as occurs in dense deposit disease (DDD), in which ribbon-like deposits replace the basement membrane (see [Chapter 23](#)). Some antigens may deposit in

TABLE 17.1 Antibodies Used in Nephropathology Diagnostics

Immunostaining	Example of Diagnostic Use
Standard	
IgG	Diseases with IgG deposits (e.g., MPGN, anti-GBM GN)
IgA	IgA nephropathy and vasculitis
IgM	Mixed cryoglobulinemia or Waldenstrom macroglobulinemia
C3	C3 glomerulopathies
C1q	Lupus nephritis
Specific	
Kappa and light chain	Renal involvement in monoclonal light chain diseases
Fibrinogen	Necrosis, thrombotic microangiopathies
Phospholipase A2 receptor	Primary membranous nephropathy
THSD7A, exostosin 1 and 2	PLA ₂ R negative membranous nephropathy
IgG1, IgG2, IgG3	Monoclonal IgG deposits
IgG4	Membranous nephropathy, IgG4-associated interstitial nephritis
DNAJB9	Fibrillary GN
Collagen type 4 alpha 3 and 5 chains	Alport syndrome
Specific amyloid antibodies (e.g., amyloid A, transthyretin)	Subtyping amyloidosis

Each nephropathology laboratory uses a different set of antibodies as a standard panel; the five mentioned in this table are considered a “minimal” standard for nontransplant kidney biopsies.

In kidney transplant biopsies, C4d and eventually stains for polyomavirus are used as a standard set, additionally, staining for adenovirus or cytomegalovirus can be used.

DNAJB9, DnaJ homolog subfamily B member 9; *GBM*, glomerular basement membrane; *GN*, glomerulonephritis; *Ig*, immunoglobulin; *MPGN*, membranoproliferative GN pattern; *PLA₂R*, phospholipase A₂ receptor; *THSD7A*, thrombospondin type 1 domain-containing protein 7A.

glomeruli and directly activate the alternative pathway of complement in the absence of IgG, as may occur in poststreptococcal GN (PSGN). Immunoglobulin with aberrant characteristics may also aggregate in glomeruli and activate complement in the absence of antigen, as occurs in IgA nephropathy with aberrantly glycosylated IgA (see [Chapter 24](#)) and glomerular diseases associated with monoclonal immunoglobulins (see [Chapters 22 and 28](#)).⁸

Normally, immune complexes are removed from the circulation by binding of the complex to the C3b receptors on erythrocytes. The immune complexes are then removed and degraded during transit of the erythrocytes in the liver and spleen. If antigenemia persists or clearance of complexes is impaired (e.g., chronic liver disease), immune complexes may deposit in the glomerulus by binding to Fc receptors on mesangial cells or by passive deposition in the mesangium or sub-endothelial space. Physical characteristics of the complexes also may favor deposition, including avidity, charge, and size. However, quantification of circulating immune complexes in patients with GN does not correlate reliably with glomerular events and thus is not typically measured.

In some glomerular diseases, the target antigen has been identified ([Table 17.2](#)). In other patients, glomerular disease develops as a result

of infection with organisms that release superantigens that cause a polyclonal activation of B cells. The classic organism responsible for superantigen-associated GN is *Staphylococcus aureus*, and the pattern of immune deposits often includes the presence of both IgG and IgA. Some infections initiate an immune response that cross-reacts with endogenous antigens. This type of molecular mimicry may be responsible for Goodpasture disease and certain types of vasculitis⁹ (see [Table 17.2](#)). Once an immune response is initiated, local injury may lead to the release of additional antigens that extend the immune response (epitope spreading). In Goodpasture disease, in which the antigen is an $\alpha 3$ chain of type IV collagen, the antigen is present in the lung alveolar basement membrane but is normally sequestered. In tobacco smokers, however, the inhalation results in oxidative injury with exposure of the $\alpha 3$ chain, allowing the binding of antibodies.¹⁰ This may explain why lung involvement rarely occurs in nonsmokers with Goodpasture disease.

Complement

The complement system is often activated in glomerular disease ([Fig. 17.6](#)). Complement can be activated through three pathways. *Classic pathway* activation involves the binding of C1q to the Fc region of antibody in IgG- and IgM-containing immune complexes and can result in reduced serum C4 and C3. This is common in lupus nephritis, immune complex-induced GN, and cryoglobulinemic MPGN. Complement may be activated by the *alternative pathway*, which is activated independently of immune complexes and can be triggered by polysaccharide antigens, polymeric IgA, injured cells, bacterial products (e.g., streptococcal antigens), and antibodies to complement pathway components (C3 convertase). The alternative pathway appears to be activated in IgA nephropathy, DDD, and PSGN. Serum complement levels are generally normal in IgA nephropathy; in DDD and PSGN, however, the C3 is typically low but C4 is normal. In DDD as well as in C3-glomerulonephritis (C3GN) (see [Chapter 23](#)), the activation of the alternative pathway may not involve an antigen but rather results from continuous activation of the pathway through altered factor H or an IgG autoantibody (nephritic factor) that stabilizes the C3 convertase (see [Chapter 23](#)). Complement also can be activated through the mannose-binding lectin (MBL) pathway initiated by MBL, which has a structure similar to that of C1q. The role of the MBL pathway in GN is emerging in IgA nephropathy and idiopathic MN. However, despite evidence for intraglomerular complement activation in these conditions, serum complement components such as C3 and C4 generally remain in the normal range.

Activation of the complement pathway has several consequences. Leukocyte recruitment is facilitated by the chemotactic factor C5a, and C3b binding is important in the binding and opsonization of the immune complexes by the infiltrating leukocytes. The terminal membrane attack complex of the cascade, C5b-9, inserts into cell membranes, where it can kill cells or more commonly activate them to secrete cytokines, oxidants, and extracellular matrix. C5b-9 likely has a role in mediating injury to the glomerular epithelial cell in MN, in which immune deposits and complement activation occur in the subepithelial space. Complement also can be activated in proteinuric urine because of amidation of C3 by ammonia, which may have a role in mediating tubulointerstitial injury even in conditions not associated with immune complex formation. Experimental studies have emphasized the importance of local synthesis of complement components by the tubular cells as a mechanism that may augment this process.¹¹

Activation of complement is controlled by complement regulatory proteins (see [Fig. 17.6](#)). A genetic absence or malfunction of factor H or other regulatory proteins can result in increased susceptibility to glomerular endothelial injury, resulting in thrombotic microangiopathy ([Chapter 30](#)) or less often hereditary forms of GN (see [Chapter 23](#)).

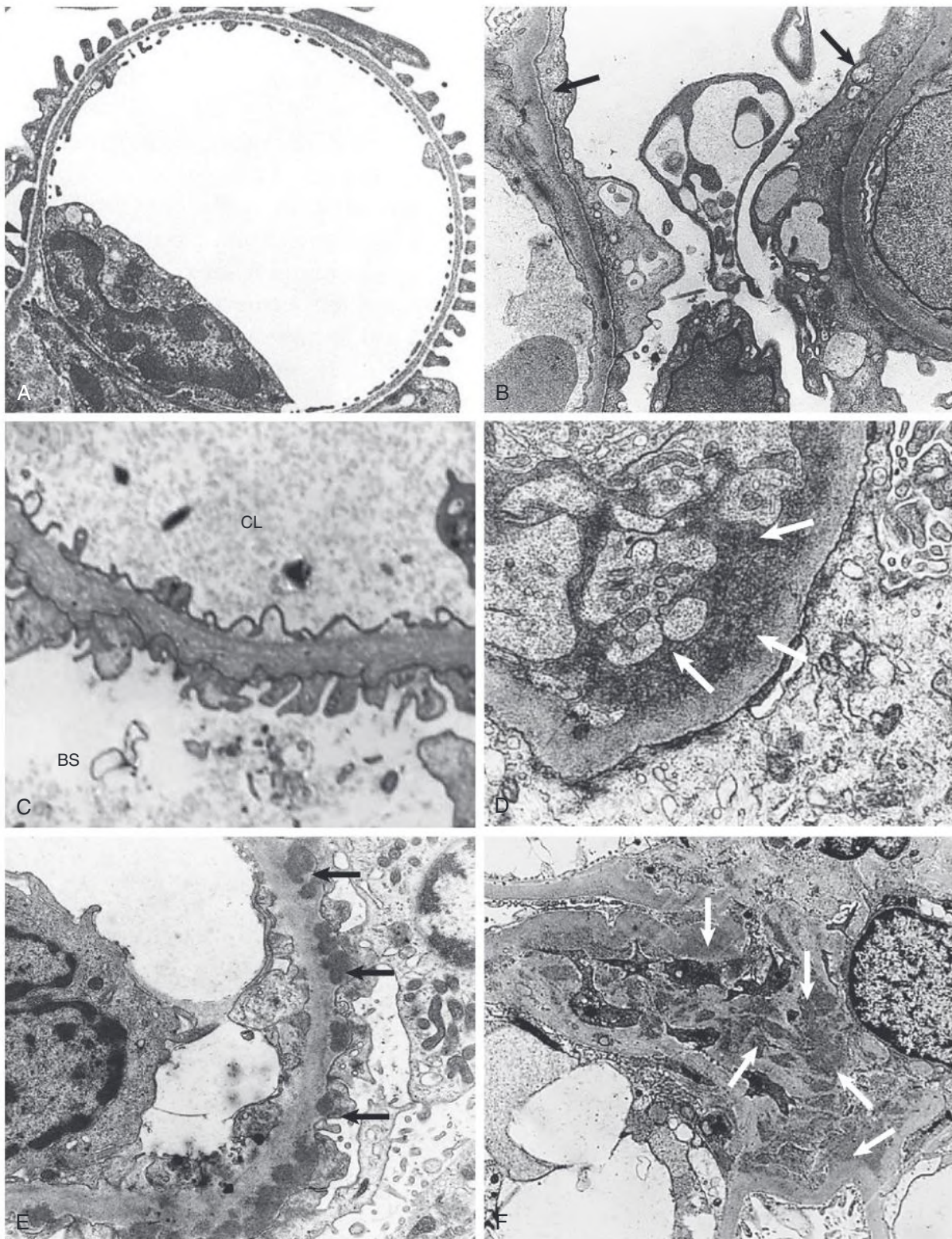


Fig. 17.3 Ultrastructural Pathology of Glomerular Disease. Some characteristic patterns of electron-dense deposits and glomerular basement membrane (GBM) abnormalities seen in glomerular disease. (A) Normal. (B) Foot process effacement: minimal change disease (*arrows*). (C) GBM thickening and splitting: Alport syndrome. (D) Subendothelial electron-dense deposits (*arrows*): membranoproliferative glomerulonephritis type I. (E) Subepithelial electron-dense deposits (*arrows*): membranous nephropathy. (F) Mesangial electron-dense deposits (*arrows*): IgA nephropathy. BS, Bowman's space; CL, capillary lumen.

Mechanisms of Proteinuria

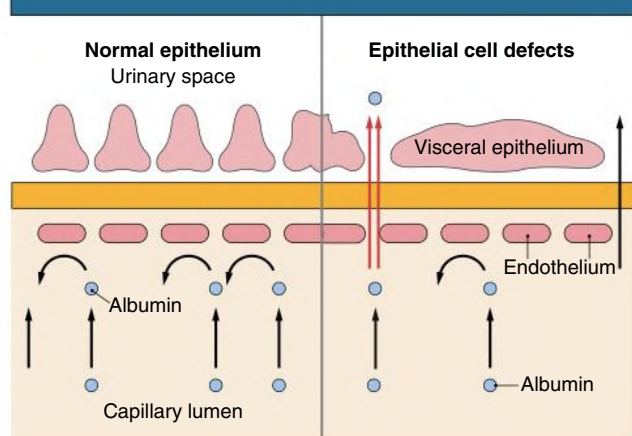


Fig. 17.4 Mechanisms of Proteinuria. Normally, negatively charged proteins such as albumin (blue circles) are repelled by the negatively charged proteins in the endothelium (sialoglycoproteins) and basement membrane (heparan sulfate proteoglycans), as well as by a size barrier in the glomerular basement membrane (GBM) and at the slit diaphragm so that only small amounts of albumin pass into the urinary space. In most proteinuric states, the podocytes are injured, leading to foot process swelling and injury to the slit diaphragm; in these situations, large amounts of protein (albumin) can pass through the GBM and the gaps between the fused foot processes (red arrows).

Glomerular Permeability in Nephrotic Syndrome

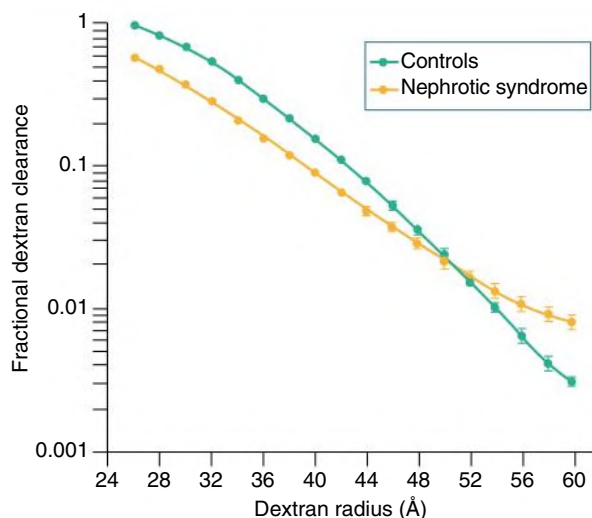


Fig. 17.5 Glomerular Permeability in Nephrotic Syndrome. Dextran sieving curve shows the relative glomerular permeability of different-sized dextrans in normal individuals and nephrotic patients with membranous nephropathy and minimal change disease. Nephrotic patients actually have a lower fractional dextran clearance for small dextrans (26–48 Å [2.6–4.8 nm]) but have an increased clearance for dextrans of larger molecular weight (52–60 Å [5.2–6.0 nm]). This is consistent with large pores appearing in the glomerular basement membrane. (Modified from Machuca E, Hummel A, Nevo F, et al. Clinical and epidemiological assessment of steroid-resistant nephrotic syndrome associated with the NPHS2 R229Q variant. *Kidney Int.* 2009;75:727–735.)

TABLE 17.2 Antigens Identified in Glomerulonephritis

Disease	Antigens
Poststreptococcal GN	Streptococcal pyrogenic exotoxin B (SPEB), plasmin receptor
Anti-GBM disease	$\alpha 3$ type IV collagen (likely induced by molecular mimicry)
IgA nephropathy	Possibly no antigen but rather polymeric polyclonal IgA (? superantigen driven)
MN ^a	Phospholipase A ₂ receptor Thrombospondin type-1 domain-containing 7A Neutral endopeptidase in podocyte (congenital) HBeAg (hepatitis B virus associated)
<i>Staphylococcus aureus</i> -associated GN	<i>Staphylococcus</i> superantigens induce polyclonal response; not necessarily antigen in glomeruli
MPGN	HCV and HBeAg in hepatitis-associated MPGN

^aSee Chapter 21 for less common newer autoantigens.

GBM, Glomerular basement membrane; GN, glomerulonephritis; HBeAg, hepatitis B early antigen; HBsAg, hepatitis B surface antigen; HCV, hepatitis C virus; IgA, immunoglobulin A; MPGN, membranoproliferative GN pattern; MN, membranous nephropathy; SPEB, streptococcal pyrogenic exotoxin B.

Mechanisms of Immune Glomerular Injury

Two major mechanisms account for the presence of immune complexes in glomerular diseases. There may be ineffectual clearance of an antigen from an impaired immune response, as in chronic viral infections caused by hepatitis B or hepatitis C virus (HBV or HCV). Despite a strong humoral response, viral infection persists because the cell-mediated response required for elimination of these viruses is impaired. The consequence is a state of persistent antigenemia with circulating antigen-antibody complexes, which predisposes to glomerular injury. Eradication of the virus with antiviral therapy can be associated with remission of the glomerular disease.

More often, glomerular disease results from autoimmunity. In health, a tension exists between the normal immune response to foreign antigen and tolerance, which is the cellular process that prevents an immune response to self-antigen. Tolerance develops because self-reactive T and B cells are clonally deleted during fetal and neonatal life, although small numbers survive outside the thymus or bone marrow, respectively. Under certain conditions, these peripheral self-reactive cells can be stimulated to generate a cellular and humoral response to a self-antigen. Infection or toxins may play a role in initiating the response by releasing antigens from sequestered sites so they have access to dendritic cells, which carry the antigen to lymph nodes for presentation to T cells, by altering host proteins to make them more immunogenic, or by molecular mimicry, in which antibodies to an exogenous antigen (e.g., those present in infecting organism) cross-react with a native protein.¹² Activation of T cells may be further enhanced by the release of cytokines or endogenous danger-associated molecular patterns, such as toll-like receptor ligands or inflammasome activators, and the conversion of normally innocuous endogenous renal cells into antigen-presenting cells through the upregulated or de novo expression of human leukocyte antigen (HLA) class II molecules and cytokines.

Regulatory T cells (CD4⁺, CD25⁺) have a key role in controlling T cell responses and preventing the development of autoimmunity. However, their role is not well understood, and the local regulation of these cells in the kidney by a subset of dendritic cells (CD103⁺) may be more important than systemic numbers of regulatory T cells.¹³

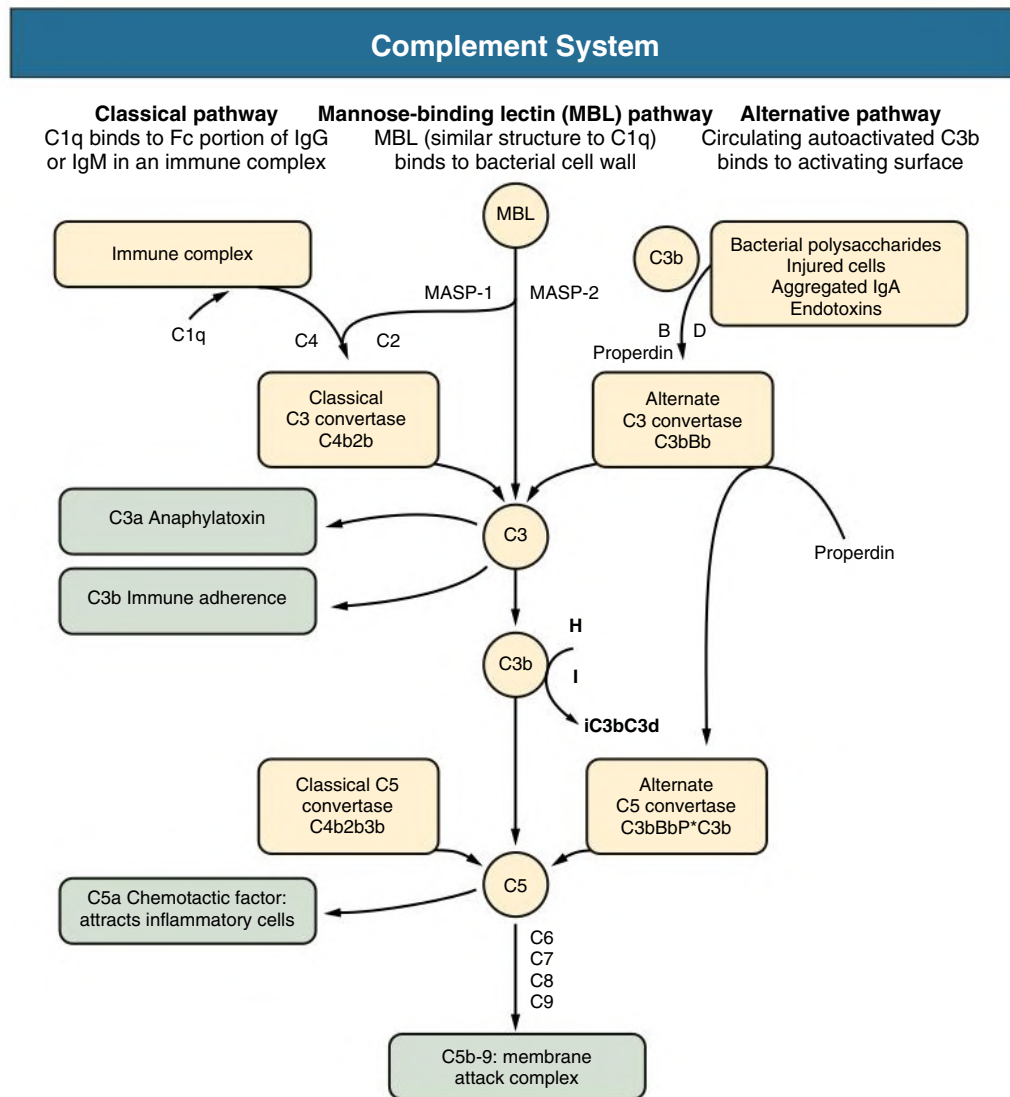


Fig. 17.6 Complement System. The complement system is a self-amplifying cascade of proteins that generates a membrane attack complex, which is cytolytic; the cascade promotes inflammation by the activity of the fragments it produces. The amplifying cascades result from activated fragments of the components combining to make convertase enzymes that degrade C3 and C5. The complement cascade is controlled in part by the short active life of many of its components. There are also inhibitory regulatory proteins, most notably factors H and I inhibiting C3b. Activated fragments of any component are designated *b* (e.g., C3b); anaphylatoxic fragments are designated *a* (e.g., C5a). Inflammatory functions of complement components are shown in green. *IgA*, Immunoglobulin A; *MASP*, MBL-associated serine protease.

Variations in HLA molecules and the T cell receptor are under strong genetic influence. Close immunogenetic associations, particularly between HLA expression and various patterns of GN, have been described in IgA nephropathy, MN, and other glomerular diseases. For example, whereas HLA-DR2 identifies a powerful relative risk for the development of Goodpasture disease, some individuals can develop the disease without HLA-DR2, and the vast majority with HLA-DR2 never develop this rare disease. HLA associations also differ among various ethnic groups. To date, HLA associations have no practical diagnostic or therapeutic implications, and HLA typing is not needed in the clinical management of patients with GN.

Inflammation

The presence of glomerular inflammation is largely determined by the site of immune deposits. Immune deposits with direct access to the circulation (subendothelial and intrabasement membrane

locations) are usually associated with leukocyte accumulation. Mesangial deposits elicit an intermediate response, whereas immune deposits in the subepithelial space generally are not associated with inflammatory cells.

In GN associated with subendothelial deposits, such as class III or IV lupus nephritis or MPGN, leukocyte infiltration is common. With acute injury, the predominant infiltrating cells are platelets and monocytes, and occasional neutrophils, and in chronic injury the predominant cells are monocyte/macrophages and T cells. The primary mechanism for attracting these cells is the secretion of chemokines and the expression of leukocyte adhesion molecules by local endothelial and resident cells; local release of complement activation fragments (C5a) is also important.

Although neutrophils are common with immune complex disease, cell-mediated immunity is important in some glomerular diseases. For example, T cells likely have a role in crescentic nephritis, becoming

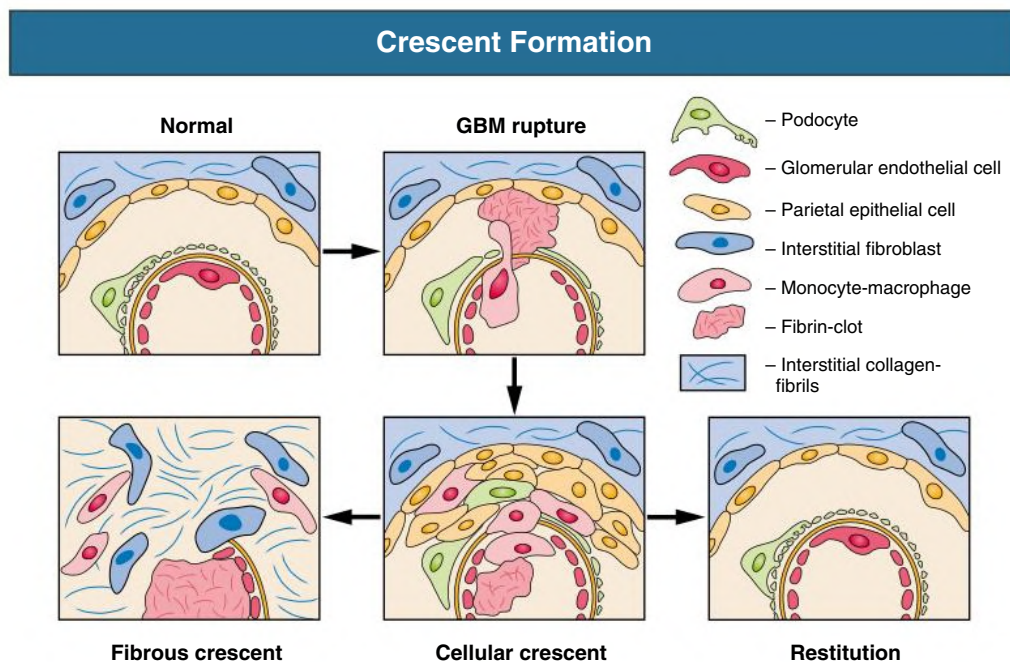


Fig. 17.7 Crescent Formation. In early crescent formation, cytokines and growth factors cross the glomerular basement membrane (GBM) to initiate proliferation of the parietal epithelial cells. Small breaks in the GBM occur secondary to injury from oxidants and proteases from neutrophils and macrophages, thus allowing the macrophage to enter Bowman's space, where it can proliferate. Breaks in Bowman's capsule secondary to the periglomerular inflammation also occur, allowing the entrance of more inflammatory cells, as well as fibroblasts. The proliferation of parietal and visceral epithelial cells and macrophages is associated with fibrin deposition, slowly choking the glomerular tuft until filtration becomes impossible. In the late stages, the crescent becomes fibrotic and the glomerulus ends stage. Alternatively, in less severe cases, complete restitution of the glomerular tuft can occur.

sensitized to endogenous or exogenous antigen and then recruiting macrophages that mediate crescent formation.

Proliferation, Apoptosis, and Fibrosis

Intrinsic glomerular cells (epithelial, mesangial, and endothelial) are also activated in various glomerular diseases. Mesangial cells can become myofibroblast-like cells that proliferate and produce excessive extracellular matrix. Endothelial cells produce nitric oxide and other anti-inflammatory proteins, and injury to this cell population can result in the expression of leukocyte adhesion molecules and activation of the coagulation system. Podocytes are differentiated epithelial cells that when injured undergo shape change (reorganization of the actin filaments) that can lead to disruption of the slit diaphragm, resulting in proteinuria. Interestingly, podocytes also can be induced to express receptors involved in antigen presentation that are similar to those expressed by dendritic cells. Progressive loss of podocytes by apoptosis is associated with the development of glomerulosclerosis. Important growth factors associated with glomerular injury include transforming growth factor β (TGF- β), which mediates matrix deposition; platelet-derived growth factor (PDGF), which mediates mesangial cell proliferation; and vascular endothelial growth factor (VEGF), required for endothelial health.

Crescent formation represents a severe cellular response and is initiated by cytokine-driven proliferation, particularly of the parietal epithelial cells. Local breaks in the GBM or Bowman's capsule, mediated by activated leukocytes, are followed by macrophage infiltration, proliferation of parietal epithelial cells and podocytes, and local fibrin deposition (Fig. 17.7).

Glomerular scarring (i.e., sclerosis) is characterized by proliferation of mesangial cells with loss (apoptosis) of endothelial cells and podocytes and deposition of extracellular matrix. Tubulointerstitial fibrosis

also accompanies progressive glomerular disease and correlates with both renal function and prognosis. Proteinuria has been shown to activate tubular cells and induce toxicity, either directly or through the generation of oxidants (from iron proteins excreted in urine) or from complement activation, which can be shown in proteinuric urine. Tubulointerstitial ischemia after loss of glomerular and peritubular capillaries also may drive fibrosis. Finally, loss of renal function may result from leakage of plasma ultrafiltrate into the peritubular space, resulting in a scarring response (misdirected filtration) or stenosis/occlusion of the opening of the proximal tubule from Bowman's space, resulting in nonfunctional (atubular) glomeruli. A detailed discussion of current mechanisms involved in glomerulosclerosis is presented in Chapter 81. Specific pathogenic mechanisms in the different patterns of glomerular disease are discussed in subsequent chapters in Section IV.

PATHOGENESIS OF SPECIFIC GLOMERULAR SYNDROMES

Minimal Change Disease

Minimal change disease (MCD) is a steroid-sensitive nephrotic syndrome in which the only structural abnormality is podocyte swelling and fusion of foot processes on EM (see Chapter 18). For many years, the podocyte injury in MCD was thought to be caused by a cytokine released from T cells. T cells are activated in MCD, and T cell hybridomas from these patients were reported to secrete a factor that provokes heavy proteinuria in rats.¹⁴ One candidate cytokine is interleukin-13 (IL-13), which is expressed by T cells in patients with MCD; overexpression of IL-13 causes nephrotic syndrome and histologic changes consistent with MCD in rats. However, proteinuria can

be induced in immunodeficient mice using CD34-positive hematopoietic bone marrow cells of patients with MCD and recurrent FSGS but not by their T cells.¹⁵ Thus, the role of T cells in this disorder remains to be clarified.

Evidence also suggests that the podocyte injury is associated with overexpression of angiotensin-like-4, which is associated with a proteinuric response.¹⁶ This overexpression can be reduced with corticosteroids and *N*-acetyl-D-mannosamine. In addition, patients with MCD show high levels of CD80 (also known as B7.1) in urine and in podocytes, and the level of urinary CD80 correlates with disease activity. CD80 is an antigen that is normally expressed by dendritic cells and B cells.

Finally, two additional discoveries have been recently made. The first is the observation that a subset of patients with MCD have antibodies to nephrin despite IF showing minimal IgG deposition in glomeruli. The other observation is that the majority of individuals with MCD have evidence for mild glomerular endothelial injury with circulating endothelial biomarkers. This has suggested that MCD is not a strict “podocytopathy” as originally proposed.

Focal Segmental Glomerulosclerosis

Focal segmental glomerulosclerosis (FSGS) is a generic term to describe a pattern of glomerular scarring and is therefore nonspecific. Focal segmental sclerosing lesions in the absence of nephrotic syndrome, often termed *secondary FSGS*, can occur in a range of contexts, including GN, chronic hypertension, obesity, and progressive renal disease of any etiology. These lesions are particularly common in African Americans; recent findings suggest this susceptibility relates to increased frequency of a genetic polymorphism in *APOL1*, a gene coding for a circulating lipoprotein.

When the histologic pattern of FSGS is seen in association with the nephrotic syndrome (a podocytopathy often termed *primary FSGS* if there is no associated condition), immune deposits are absent, but as with MCD, there is a generalized foot process fusion (also termed *effacement*) on EM (see Chapter 19). However, unlike MCD, there is segmental scarring (sclerosis) in some glomeruli with FSGS. Some forms of FSGS result from mutations of podocyte proteins (see Chapter 20), whereas others are believed to originate from a circulating factor; cardiotrophin-like cytokine 1 is one candidate molecule, but others are also being investigated as potential circulating factors that may mediate FSGS (see Chapter 19). Although it is unproven, some consider that MCD and FSGS share similar pathogenic mechanisms and are part of a “podocytopathy” spectrum in which MCD has lower levels of the circulating factor and is therefore more sensitive to corticosteroids. These forms of FSGS may be particularly prone to rapid recurrence, sometimes within hours, after renal transplantation.

A variant of FSGS is *collapsing FSGS*, in which there is proliferation of the normally quiescent podocyte, leading to collapse of the glomerular tuft, often in association with massive proteinuria. The pathogenesis may involve production by the podocyte of growth factors such as VEGF or local inhibition of cell cycle proteins that normally maintain the podocyte in a nonproliferative state.¹⁷

Membranous Nephropathy

In membranous nephropathy (MN), immune deposits are localized to the subepithelial space, representing autoantibody binding to an intrinsic podocyte antigen (see Chapter 21). The major antigen is the M-type phospholipase A₂ receptor (PLA₂R), accounting for as many as 70% of cases of idiopathic MN.¹⁸ Antibodies to PLA₂R are specific for MN. Circulating anti-PLA₂R reflects the immunologic activity of the disease and may be useful to monitor the clinical course, including patient response to treatment. Other autoantibodies have been found, albeit much less frequently, including those against thrombospondin type-1 domain-containing 7A,¹⁹ exostosin-1 and exostosin-2,²⁰ neutral

endopeptidase (in the very rare neonatal MN), or against bovine serum albumin (BSA).²¹ Some cases of MN may be caused by low-avidity immune complexes, which may dissociate and then reform at the subepithelial space; this may be a mechanism for some MN caused by viruses such as HBV.

Most cases of idiopathic MN caused by anti-PLA₂ antibodies are associated with IgG4 deposition, which is an isoform of IgG that does not activate the classic complement pathway. However, there is evidence that the IgG4-PLA₂ complex can activate complement through the MBL complement pathway, resulting in local generation of the membrane attack complex (C5b-9), which may insert into the podocyte to cause activation, injury, and proteinuria.

Glomerulonephritis With a Membranoproliferative Pattern on Histology

The term MPGN describes a histologic pattern rather than a single disease entity. The pattern may result from localization of immune deposits to both the mesangium and the subendothelial space (see Chapter 22). One of the diseases causing an MPGN pattern is cryoglobulinemic GN, in which the immune complexes contain a monoclonal IgM or polyclonal IgM that acts as a rheumatoid factor by binding to the IgG in the immune complex. Whether due to cryoglobulinemia or another cause, the MPGN pattern and its clinical consequences are thought to occur by passive deposition from the circulation, and the antigen is often a component of the HCV virus, especially in adults. When this pattern is seen in lupus nephritis, it may be facilitated by the binding of extracellular nucleosomes to the complexes. Nucleosomes are cationic nuclear proteins that can interact with the negatively charged proteins within the glomerulus.

Studies in experimental models suggest that the intraglomerular immune complexes cause local complement activation with the generation of chemotactic factors, including C5a, chemokines, and leukotrienes. Leukocyte adhesion molecules on endothelial cells are upregulated (intracellular adhesion molecule 1) or expressed *de novo* (E- and P-selectin). Proinflammatory cytokines (IL-1 and tumor necrosis factor- α) are generated locally and augment the inflammatory response. Neutrophils, platelets, and monocytes/macrophages then localize in the glomerulus and release oxidants, particularly hypohalous acids generated by neutrophil myeloperoxidase, and proteases (elastase, cathepsin G, metalloproteinases) that cause local cellular injury and GBM degradation.

Dense Deposit Disease and C3 Glomerulonephritis

In contrast to the MPGN forms described above, immune complexes are absent in glomeruli of patients with DDD and C3-GN. Initiation results from spontaneous intraglomerular activation of the alternative complement pathway. The most common cause is nephritic factor, an autoantibody that activates the alternative pathway. Some cases may be caused by mutations of the complement regulatory factor H, in which case the location of the mutation within the gene determines whether the disease manifests as DDD or atypical HUS.²²

Mesangial Proliferative Glomerulonephritis

IgA nephropathy is the most common type of GN (see Chapter 24). Production of an abnormally glycosylated IgA, possibly by a bacterial superantigen,²³ or perhaps from an altered mucosal immune system, may lead to IgA polymers that deposit in the mesangium; the glomerular capillary wall is relatively spared. Marked yet usually nonnephrotic proteinuria is a common feature of the clinical presentation. Mesangial cell injury may be mediated by binding of the IgA-containing immune complexes to Fc α or other IgA receptors on the mesangial cell, resulting in the release of chemokines and growth factors that provokes leukocyte infiltration as well as mesangial cell proliferation and mesangial matrix production.

Poststreptococcal Glomerulonephritis

Poststreptococcal GN (PSGN) has long been considered the human equivalent of acute serum sickness in rabbits (see [Chapter 57](#)). It is observed only in patients infected with specific (nephritogenic strains) of group A streptococci. One antigen responsible for some cases of PSGN, streptococcal pyrogenic exotoxin B, enters the circulation and localizes to glomeruli, resulting in a brisk inflammatory reaction with local endothelial and mesangial cell proliferation and manifestations of nephritic syndrome. Complement activation occurs through the alternative pathway and may result from direct activation of the pathway by streptococcal antigens. Some deposits (“humps”) also form in the subepithelial space and may represent the translocation of immune complexes across the GBM or in situ immune complex formation.

Goodpasture Disease

Goodpasture disease (anti-GBM disease) is caused by an autoantibody to the $\alpha 3$ chain of type IV collagen present in the GBM and alveolar basement membrane (see [Chapter 25](#)). The autoantibody develops in genetically susceptible individuals because of molecular mimicry between the type IV collagen antigens and certain bacterial antigens.⁹ Binding of antibody results in complement activation with the infiltration of inflammatory cells, causing local capillary wall damage and

proteinuria. Crescent formation also usually occurs and may be mediated by both T cells and macrophages.

Antineutrophil Cytoplasmic Antibody–Associated Vasculitis

A severe form of segmental necrotizing glomerular injury, often in association with crescents, can be observed with vasculitis (see [Chapter 26](#)). The two most common types of vasculitis causing this type of injury are granulomatosis with polyangiitis (GPA) and microscopic polyangiitis (MPA). Both are associated with circulating antibodies against neutrophil cytoplasmic antigens (ANCA), with antibodies to endopeptidase (proteinase) 3, which give a cytoplasmic pattern by staining (c-ANCA) in most patients with GPA granulomatosis, and antibodies to myeloperoxidase, which give a perinuclear staining pattern (p-ANCA) in subjects with MPA. Experimental evidence suggests that ANCAs are pathogenic by activating neutrophils within the vasculature. The mechanism responsible for triggering autoantibodies to neutrophil antigens remains unclear, although geographic and temporal clustering suggest a role for infection or antigen exposure that may induce an autoimmune response.¹²

Further discussion of specific pathogenic mechanisms in the different patterns of glomerular disease can be found in [Chapters 18 to 30](#).

SELF-ASSESSMENT QUESTIONS

- MN may result from all of the following *except*:
 - Anti-phospholipase A₂ receptor antibodies
 - Thrombospondin type-1 domain-containing 7A antibodies
 - Antibodies to cationic bovine serum albumin
 - Factor H deficiency
 - Deposition of hepatitis B virus antigens and antibody
- A low serum C3 level would suggest all the following diseases *except*:
 - Membranoproliferative glomerulonephritis secondary to hepatitis C virus
 - Cryoglobulinemic glomerulonephritis accompanying bacterial endocarditis
 - Class IV lupus nephritis
 - Goodpasture disease
 - Class III lupus nephritis
- Marked proteinuria typically results from damage to which structures?
 - Mesangial cells
 - Parietal epithelial cells
 - Podocytes
 - Preglomerular arterioles
 - Loops of Henle
- Which statement best describes FSGS?
 - It is typically a disease of the mesangium.
 - It usually results in nephrotic syndrome.
 - It can result from a circulating factor.
 - It is an autoimmune disease.
 - Most cases result from mutations.

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